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BIOCHIP AND MANUFACTURING METHOD THEREOF

Abstract

A biochip is configured for detecting biological materials in a solution and includes at least one detection unit, which includes a substrate, a first insulation layer disposed on the substrate, a semiconductor layer, a second insulation layer, a first metal layer, a second metal layer and a protection layer. The semiconductor layer is disposed on the first insulation layer and includes multiple reaction regions. The second insulation layer is disposed on the semiconductor layer and includes a first part, a second part surrounding the first part and a first opening. The first opening exposes the reaction regions. The first metal layer is disposed on the second part and includes source electrode, drain electrode, gate electrode and first wall structure. The protection layer includes a flat part defining a second opening, a first protruding part defining a third opening and a second protruding part defining a fourth opening.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the priority benefit of Taiwan application serial no. 113107268, filed on Feb. 29, 2024. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

[0002] The disclosure relates to a semiconductor chip and a manufacturing method thereof, and in particular relates to a biochip and a manufacturing method thereof.

Description of Related Art

[0003] In a general biochip, the space that may accommodate solution is usually limited by the size of the reaction region. Therefore, once the solution is in a larger volume or when errors occur in the addition of the solution, the problem of solution overflow may easily occur.

SUMMARY

[0004] A biochip and a manufacturing method thereof, which may prevent the issue of solution overflow and accommodate a larger volume of solution, enhancing the sensitivity of detection and simultaneously detecting various biological materials without interfering with each other, are provided in the disclosure.

[0005] The biochip of the disclosure is configured for detecting biological materials in a solution and includes at least one detection unit. The detection unit includes a substrate, a first insulation layer, a semiconductor layer, a second insulation layer, a first metal layer, a second metal layer, and a protection layer. The first insulation layer is disposed on the substrate. The semiconductor layer is disposed on the first insulation layer and includes multiple reaction regions. The second insulation layer is disposed on the semiconductor layer and includes a first part, a second part, and a first opening. The second part surrounds the first part. The first opening separates the first part and the second part and exposes the reaction regions. The first metal layer is disposed on the second part and includes a source electrode, a drain electrode, a gate electrode and a first wall structure that are separated from each other. The second metal layer includes a second wall structure and a biological electrode. The second wall structure is disposed on the first wall structure, and the biological electrode is disposed on the first part. The protection layer is disposed on the first metal layer and the second metal layer. The protection layer has a second opening, a third opening and a fourth opening, and includes a flat part, a first protruding part and a second protruding part. The flat part covers the second part exposed by the first metal layer and surrounds and defines the second opening. The first protruding part covers the source electrode, the drain electrode and the gate electrode, and surrounds and defines the third opening. The second protruding part covers the second wall structure and surrounds and defines the fourth opening. In a normal direction of the substrate, the first opening, the second opening, the third opening, and the fourth opening overlap.

[0006] In an embodiment of the disclosure, the semiconductor layer further includes a source region and a drain region. The source electrode surrounds the first opening and is electrically connected to the source region. The drain electrode surrounds the source electrode and is electrically connected to the drain region, the gate electrode surrounds the drain electrode and is electrically connected to the substrate, and the first wall structure surrounds the gate electrode.

[0007] In an embodiment of the disclosure, the source electrode, the drain electrode, the gate

electrode and the first wall structure are on the same layer, and the first wall structure electrically insulates the source electrode, the drain electrode and the gate electrode.

[0008] In an embodiment of the disclosure, the solution is disposed in at least the first opening, the second opening and the third opening. The solution contacts the biological electrode and multiple reaction regions of the semiconductor layer.

[0009] In one embodiment of the disclosure, the biological electrode and the second wall structure are on the same layer, the biological electrode and the second wall structure are separated from each other, and the second wall structure electrically insulates the biological electrode.

[0010] In an embodiment of the disclosure, the first protruding part completely surrounds the third opening, and the second protruding part completely surrounds the fourth opening.

[0011] In an embodiment of the disclosure, in the normal direction, an upper surface of the second protruding part is higher than an upper surface of the first protruding part, and the upper surface of the first protruding part is higher than an upper surface of the flat part.

[0012] In an embodiment of the disclosure, the third opening is larger than the second opening, and the fourth opening is larger than the third opening.

[0013] In an embodiment of the disclosure, the third opening includes an adding region and multiple detection regions. The adding region is disposed corresponding to the first part. The detection regions are connected to the adding region and are disposed corresponding to multiple reaction regions.

[0014] The manufacturing method of the biochip of the disclosure includes the following operation. A substrate is provided. A first insulation layer is formed on the substrate. A semiconductor layer is formed on the first insulation layer, in which the semiconductor layer includes multiple reaction regions. A second insulation layer is formed on the semiconductor layer, in which the second insulation layer includes a first part, a second part and a first opening, the second part surrounds the first part, and the first opening separates the first part and the second part and exposes multiple reaction regions. A first metal layer is formed on the second part, in which the first metal layer includes a source electrode, a drain electrode, a gate electrode and a first wall structure that are separated from each other. A second metal layer is formed, in which the second metal layer includes a second wall structure and a biological electrode, the second wall structure is disposed on the first wall structure, and the biological electrode is disposed on the first part. A protection layer is formed on the first metal layer and the second metal layer, in which the protection layer has a second opening, a third opening and a fourth opening, and the protection layer includes a flat part, a first protruding part and a second protruding part. The flat part covers the second part exposed by the first metal layer and surrounds and defines the second opening. The first protruding part covers the source electrode, the drain electrode and the gate electrode, and surrounds and defines the third opening. The second protruding part covers the second wall structure and surrounds and defines the fourth opening. In a normal direction of the substrate, the first opening, the second opening, the third opening, and the fourth opening overlap.

[0015] Based on the above, in the biochip and the manufacturing method thereof according to an embodiment of the disclosure, by disposing multiple reaction regions in one detection unit to detect the same kind of biological material, the detection sensitivity may be improved by accumulating signals detected by the multiple reaction regions. Compared with general biochips, the biochip of this embodiment may increase the volume of solution that the biochip may hold through the arrangement of the fourth opening, so as to accommodate a larger volume of solution without the concern of overflow and subsequent cross-contamination. This in turn enhances the operational flexibility and convenience of the biochip. In addition, since the multiple detection units in the biochip of this embodiment may be respectively used to detect different kinds of biological materials without the concern of overflow and subsequent cross-contamination between different detection units. This in turn enables the biochip to detect multiple biological materials at the same time without interfering with each other.

[0016] In order to make the above-mentioned features and advantages of the disclosure comprehensible, embodiments accompanied with drawings are described in detail below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] FIG. 1 is a top schematic diagram of a biochip of an embodiment of the disclosure.

[0018] FIG. 2A to FIG. 2C are top schematic diagrams of the manufacturing method of the detection unit in the biochip of FIG. 1.

[0019] FIG. 3A to FIG. 3E are three-dimensional schematic diagrams of the manufacturing method of the region R in the biochip of FIG. 2C.

[0020] FIG. 4 is a cross-sectional schematic diagram of the biochip of FIG. 2C along the section line I-I'.

[0021] FIG. 5 is a cross-sectional schematic diagram of the biochip of FIG. 2C along the section line II-II'.

[0022] FIG. 6 is a top schematic diagram of a detection unit in a biochip of another embodiment of the disclosure.

DETAILED DESCRIPTION OF DISCLOSED EMBODIMENTS

[0023] FIG. 1 is a top schematic diagram of a biochip of an embodiment of the disclosure. FIG. 2A to FIG. 2C are top schematic diagrams of the manufacturing method of the detection unit in the biochip of FIG. 1. FIG. 3A to FIG. 3E are three-dimensional schematic diagrams of the manufacturing method of the region R in the biochip of FIG. 2C. FIG. 4 is a cross-sectional schematic diagram of the biochip of FIG. 2C along the section line I-I'. FIG. 5 is a cross-sectional schematic diagram of the biochip of FIG. 2C along the section line II-II'. For clarity of the drawings and convenience of explanation, FIG. 2A to FIG. 2C omit the substrate **110**, the first insulation layer **IL1**, the second insulation layer **130**, the insulation layer **IL2**, and the protection layer **160** in the biochip **10**.

[0024] Referring to FIG. 1, FIG. 2C, FIG. 3E, FIG. 4 and FIG. 5 at the same time, the biochip **10** of this embodiment may include at least one detection unit **100** (FIG. 1 schematically illustrates three detection units **100**, but not limited thereto). The detection unit **100** includes a substrate **110**, a first insulation layer **IL1**, a semiconductor layer **120**, a second insulation layer **130**, a first metal layer **140**, a second metal layer **150**, and a protection layer **160**. The first insulation layer **IL1** is disposed on the substrate **110**. The semiconductor layer **120** is disposed on the first insulation layer **IL1** and includes multiple reaction regions **121**. The second insulation layer **130** is disposed on the semiconductor layer **120**, and the second insulation layer **130** includes a first part **131**, a second part **132**, and a first opening **O1**. The second part **132** surrounds the first part **131**. The first opening **O1** separates the first part **131** and the second part **132**, and the first opening **O1** exposes the reaction regions **121**. The first metal layer **140** is disposed on the second part **132**, and the first metal layer **140** includes a source electrode **141**, a drain electrode **142**, a gate electrode **143** and a first wall structure **144** that are separated from each other. The second metal layer **150** includes a second wall structure **152** and a biological electrode **151**. The second wall structure **152** is disposed on the first wall structure **144**, and the biological electrode **151** is disposed on the first part **131**. The protection layer **160** is disposed on the first metal layer **140** and the second metal layer **150**. The protection layer **160** has a second opening **O2**, a third opening **O3**, and a fourth opening **O4**, and the protection layer **160** includes a first protruding part **161**, a second protruding part **162**, and a flat part **163**. The flat part **163** covers the second part **132** exposed by the first metal layer **140**, and the flat part **163** surrounds and defines the second opening **O2**. The first protruding part **161** covers the source electrode **141**, the drain electrode **142** and the gate electrode **143**, and the first protruding part **161** surrounds and defines the third opening **O3**. The second protruding part **162**

covers the second wall structure **152**, and the second protruding part **162** surrounds and defines the fourth opening **O4**. In the normal direction Z of the substrate **110**, the first opening **O1**, the second opening **O2**, the third opening **O3**, and the fourth opening **O4** overlap. In addition, the biochip **10** of this embodiment may be configured for detecting the biological material **210** in the solution **200**.

[0025] The manufacturing method of the biochip **10** of this embodiment is described below. The manufacturing method of the biochip **10** in this embodiment may include the following steps.

[0026] First, referring to FIG. 2A, FIG. 3A, FIG. 4 and FIG. 5 at the same time, a substrate **110** is provided, a first insulation layer **IL1** is formed on the substrate **110**, and a semiconductor layer **120** is formed on the first insulation layer **IL1**. In this embodiment, the substrate **110** may be a silicon substrate or a silicon wafer. For example, the substrate **110** may be a P-type silicon substrate, but not limited thereto. In this embodiment, the first insulation layer **IL1** may be, for example, a gate oxide layer, but not limited thereto.

[0027] In this embodiment, the semiconductor layer **120** includes multiple reaction regions **121**, multiple source regions **122**, multiple drain regions **123**, and a central opening **124**. The source region **122** and the drain region **123** are separated from each other. The reaction region **121** is located between the adjacent source region **122** and the drain region **123**, and the reaction region **121** may connect the source region **122** and the drain region **123**. Each reaction region **121** may include at least one reaction unit **1211** (FIG. 2A schematically illustrates five reaction units **1211**, but not limited thereto). The central opening **124** may be surrounded by multiple reaction regions **121**, multiple source regions **122** and multiple drain regions **123**. In this embodiment, the material of the semiconductor layer **120** may include polysilicon or other suitable semiconductor materials, but not limited thereto. In some embodiments, the reaction unit **1211** in the reaction region **121** may be regarded as a channel in the transistor structure. Therefore, when the threshold voltage of the channel (reaction unit **1211**) is exceeded, the channel (reaction unit **1211**) may be opened, and the current from the drain electrode **142** may be transferred to the source electrode **141** through the channel (reaction unit **1211**).

[0028] In this embodiment, in the top schematic diagram of FIG. 2A, the semiconductor layer **120** may schematically include eight reaction regions **121**, four source regions **122**, four drain regions **123**, and one central opening **124**. The eight reaction regions **121**, four source regions **122**, four drain regions **123**, and one central opening **124** of the semiconductor layer **120** may be arranged into a hollow circular structure. Each pair of adjacent source regions **122** and drain regions **123** may occupy approximately 90 degrees of the entire 360-degree circular structure, and each source region **122** (or drain region **123**) may occupy approximately 45 degrees of the entire 360-degree circular structure, but not limited thereto. That is, the disclosure does not limit the number and arrangement shapes of the reaction regions, source regions, drain regions and central openings, and the disclosure does not limit the ratio of each adjacent source region and drain region and the ratio of each source (or drain) region occupying the entire circle.

[0029] In addition, in this embodiment, an identification unit (not shown) may also be provided on the reaction unit **1211** of the reaction region **121** of the semiconductor layer **120** for specificity identification and binding to the biological material **210** in the solution **200**. Specifically, one end of the identification unit may be connected and fixed to the reaction region **121**, and the other end of the identification unit may be configured for identification and is bound to the biological material **210**. The identification unit may be a chemical molecule or a biological molecule. For example, the identification unit may be an antibody, an antigen, a nucleic acid, a carbohydrate, or a combination thereof, but not limited thereto, as long as the identification unit may specifically identify and bind to the biological material **210**.

[0030] Then, referring to FIG. 2A, FIG. 3B, FIG. 4 and FIG. 5 at the same time, a second insulation layer **130** is formed on the semiconductor layer **120**. Specifically, the second insulation layer **130** may cover the semiconductor layer **120** and a part of the first insulation layer **IL1**. The second insulation layer **130** includes a first part **131**, a second part **132**, a first opening **O1**, a hole

133, a hole **134**, and a hole **135**. The first part **131** is disposed corresponding to the central opening **124**. The second part **132** and the first part **131** are separated from each other, and the second part **132** surrounds the first part **131**. The first opening **O1** separates the first part **131** and the second part **132**. The first opening **O1** may expose multiple reaction regions **121** and a part of the first insulation layer **IL1**. The hole **133** may penetrate the second part **132** and expose a part of the source region **122**, the hole **134** may penetrate the second part **132** and expose a part of the drain region **123**, and the hole **135** may penetrate the second part **132** and the first insulation layer **IL1** to expose a part of the substrate **110**.

[0031] Then, referring to FIG. 2B, FIG. 3C, FIG. 4 and FIG. 5 at the same time, a first metal layer **140** is formed on the second part **132** of the second insulation layer **130**. Specifically, the first metal layer **140** may expose the first part **131** and a part of the second part **132**. The first metal layer **140** includes a source electrode **141**, a drain electrode **142**, a gate electrode **143**, and a first wall structure **144** that are separated from each other. In the normal direction **Z** of the substrate **110**, the source electrode **141** may overlap and be disposed corresponding to the source region **122**, and the drain electrode **142** may overlap and be disposed corresponding to the drain region **123**. The source electrode **141** may be disposed on the second part **132** and in the hole **133** and be electrically connected to the source region **122**. The drain electrode **142** may be disposed on the second part **132** and in the hole **134** and be electrically connected to the drain region **123**. The gate electrode **143** may be disposed on the second part **132** and in the hole **135** to contact and be electrically connected to the substrate **110** with conductive properties.

[0032] In this embodiment, the source electrode **141**, the drain electrode **142**, the gate electrode **143**, and the first wall structure **144** may be on the same layer. As shown in FIG. 2B, the source electrode **141** may have a substantially annular structure surrounding the first opening **O1**, the drain electrode **142** may have a substantially annular structure surrounding the source electrode **141** and the first opening **O1**, the gate electrode **143** may have a substantially annular structure surrounding the drain electrode **142**, the source electrode **141**, and the first opening **O1**, and the first wall structure **144** may have a substantially annular structure surrounding the gate electrode **143**, the drain electrode **142**, the source electrode **141**, and the first opening **O1**. The first wall structure **144** may electrically insulate the source electrode **141**, the drain electrode **142**, and the gate electrode **143**.

[0033] Then, referring to FIG. 2C, FIG. 3D, FIG. 4, and FIG. 5 at the same time, an insulation layer **IL2** is formed on the first part **131** and the first wall structure **144**, and a second metal layer **150** is formed on the first part **131**, the first wall structure **144**, and the insulation layer **IL2**. Specifically, the second metal layer **150** includes a biological electrode **151** and a second wall structure **152**. The biological electrode **151** is disposed on the first part **131**. The second wall structure **152** is disposed on the first wall structure **144**. The biological electrode **151** and the second wall structure **152** may be on the same layer, and the biological electrode **151** and the second wall structure **152** are separated from each other. As shown in FIG. 2C, the second wall structure **152** may have a substantially annular structure surrounding the gate electrode **143**, the drain electrode **142**, the source electrode **141**, and the first opening **O1**. The second wall structure **152** electrically insulates the biological electrode **151**.

[0034] Then, referring to FIG. 2C, FIG. 3E, FIG. 4, and FIG. 5 at the same time, a protection layer **160** is formed on the first metal layer **140** and the second metal layer **150**. Specifically, the protection layer **160** has a second opening **O2**, a third opening **O3**, and a fourth opening **O4**, and the protection layer **160** includes a flat part **163**, a first protruding part **161**, and a second protruding part **162**. The flat part **163** may cover the second part **132** exposed by the first metal layer **140**, and the flat part **163** may surround and define the second opening **O2**. The first protruding part **161** may cover the source electrode **141**, the drain electrode **142** and the gate electrode **143**, and the first protruding part **161** may surround and define the third opening **O3**. The second protruding part **162** may cover the second wall structure **152**, and the second protruding part **162** may surround and

define the fourth opening O4.

[0035] In this embodiment, the flat part **163** has an upper surface **163a** away from the second insulation layer **130**, the first protruding part **161** has an upper surface **161a** away from the second insulation layer **130**, and the second protruding part **162** has an upper surface **162a** away from the second insulation layer **130**. In the normal direction Z of the substrate **110**, the upper surface **161a** of the first protruding part **161** is higher than the upper surface **163a** of the flat part **163**, and the upper surface **162a** of the second protruding part **162** is higher than the upper surface **161a** of the first protruding part **161**.

[0036] In this embodiment, through the arrangement of the first metal layer **140** and the second metal layer **150**, the first protruding part **161** and the second protruding part **162** may be formed simultaneously during the step of forming the protection layer **160**, so it is not necessary to have additional process steps (e.g., increasing the number of photomasks or increasing the number of layers) to manufacture the first protruding part **161** that may be used to form the third opening O3 and the second protruding part **162** that may be used to form the fourth opening O4, which has the effect of simplifying the process.

[0037] In this embodiment, in the normal direction Z of the substrate **110**, the first opening O1, the second opening O2, the third opening O3, and the fourth opening O4 may overlap. The second opening O2 may connect the first opening O1 and the third opening O3, and the third opening O3 may connect the second opening O2 and the fourth opening O4. The size of the second opening O2 may be substantially similar to the size of the first opening O1. The size of the third opening O3 may be larger than the size of the second opening O2. The size of the fourth opening O4 may be larger than the size of the third opening O3.

[0038] In this embodiment, the third opening O3 may include an adding region O31 and multiple detection regions O32. The adding region O31 may be a region where the solution **200** is added to the biochip **10**, and the detection region O32 may be a region configured to guide and store the solution **200** to avoid overflow of the solution **200**. Specifically, the adding region O31 may be disposed corresponding to the central opening **124**, the first part **131**, and the biological electrode **151**. The detection regions O32 may be disposed corresponding to the reaction regions **121**. The detection regions O32 may be connected to the adding region O31, and the detection regions O32 may be arranged radially around the adding region O31, thereby the time for the solution **200** to flow into the detection region O32 may be shortened, and the distribution of the solution **200** may be more uniform.

[0039] In this embodiment, since the solution **200** may be first dropped into the adding region O31 in the form of liquid beads, and then flow from the adding region O31 to multiple radially arranged detection regions O32, therefore, by making the diameter (or width) of the adding region O31 larger than the diameter of the liquid bead, the solution **200** may be prevented from overflowing due to the offset (or misalignment) of the adding position during addition. By designing the outline shape of the adding region O31 to be circular to match the shape of the liquid beads, the surface tension of the liquid may be used to prevent the solution **200** from overflowing when the amount of liquid in the solution **200** is too much and is close to overflowing. In other words, this embodiment utilizes the “self-confined structure” formed by the wiring of the biochip itself to increase the alignment error tolerance space and accommodate more solution **200**. In this embodiment, the diameter (or width) of the adding region O31 is, for example, about 10 micrometers (μm) to 100 micrometers, and the width of the detection region O32 is, for example, about 3 micrometers, but not limited thereto. In some embodiments, the diameter (or width) of the adding region O31 may also be adjusted based on the diameter of the liquid beads, and the length of the detection region O32 may also be adjusted based on the number of reaction units **1211**.

[0040] Then, referring to FIG. 4 and FIG. 5 at the same time, in this embodiment, the solution **200** may be disposed at least in the first opening O1, the second opening O2, and the third opening O3, and the solution **200** may at least cover the upper surface **163a** of the flat part **163**, so that the

solution **200** may contact the biological electrode **151** and the reaction units **1211** in the reaction regions **121** at the same time. In some embodiments, when the volume of solution **200** is large, the solution **200** may also be disposed in the first opening **O1**, the second opening **O2**, the third opening **O3**, and the fourth opening **O4**, so that the solution **200** may cover the upper surface **163a** of the flat part **163** and the upper surface **161a** of the first protruding part **161**.

[0041] Compared with general biochips that use external probe electrodes (e.g., silver/silver chloride electrodes) to detect biological materials, this embodiment may significantly reduce the overall volume of the biochip **10** by integrating the biological electrode **151** into the biochip **10**, and may reduce the complicated steps and costs of the post-processing process (e.g., it is not necessary to manufacture additional external probe electrodes).

[0042] In this embodiment, the solution **200** may include, for example, body fluids such as serum, and the biological material **210** may include, for example, microorganisms or biomolecules, but not limited thereto. For example, microorganisms may include bacteria, viruses, or combinations thereof, and biomolecules may include nucleic acids (including DNA, ribonucleic acids, or combinations thereof), nucleotides, proteins, carbohydrates, lipids, or combinations thereof, but not limited thereto.

[0043] In this embodiment, the working principle of the biochip **10** may include testing before the solution **200** is added and detection after the solution **200** is added. Specifically, the testing before the solution **200** is added is: the gate electrode **143** is first turned on, so that the voltage provided by the gate electrode **143** may control (turn on or off) the reaction unit **1211** in the reaction region **121** through the substrate **110** with conductive properties separated by the first insulation layer **IL1**. Then, the current amount of the drain electrode **142** (i.e., the first current amount) is measured to test whether the reaction unit **1211** may be normally conducted so that the current from the source electrode **141** may pass. The detection after the solution **200** is added is: when the solution **200** may contact the biological electrode **151** and the reaction units **1211** in the reaction regions **121** at the same time, the biological electrode **151** is first turned on, so that the voltage provided by the biological electrode **151** may control (turn on or off) the reaction unit **1211** in the reaction region **121** separated by the solution **200**. Then, the current amount of the drain electrode **142** (i.e., the second current amount) is measured, and the first current amount and the second current amount are compared so as to detect the biological material **210** in the solution **200**. In addition, in this embodiment, during the detection after the solution **200** is added, the gate electrode **143** may be in an off or on state. When the gate electrode **143** is in an on state, the voltage provided by the gate electrode **143** can, for example, be used to enhance the reaction between the biological material **210** and the reaction unit **1211**.

[0044] In this embodiment, the reaction regions **121** in one of the detection units **100** of the biochip **10** may be configured to detect the same kind of biological material **210**, thereby the detection sensitivity may be improved by accumulating signals detected by the reaction regions **121**.

[0045] In this embodiment, the first protruding part **161** may completely surround the third opening **O3**, and the second protruding part **162** may completely surround the fourth opening **O4**. In detail, since the source electrode **141**, the drain electrode **142**, and the gate electrode **143** have substantially annular structures that may surround the first opening **O1**, the first protruding part **161** disposed above the source electrode **141**, the drain electrode **142** and the gate electrode **143** may be a closed pattern that completely surrounds the first opening **O1**, the second opening **O2**, and the third opening **O3**, so as to confine the solution **200** within the third opening **O3** and prevent the solution **200** from overflowing outside the third opening **O3**. In addition, since the second wall structure **152** is disposed on the first wall structure **144** and the second wall structure **152** has substantially an annular structure that may surround the first opening **O1**, the second protruding part **162** disposed above the second wall structure **152** may be a closed pattern that completely surrounds the first opening **O1**, the second opening **O2**, the third opening **O3**, and the fourth opening **O4**, so as to confine the solution **200** within the fourth opening **O4** and prevent the

solution **200** from overflowing outside the fourth opening **O4**, as shown in FIG. **4** and FIG. **5**.
[0046] For example, when the solution **200** added to the third opening **O3** overflows, the arrangement of the second protruding part **162** may confine the solution **200** within the fourth opening **O4** and prevent the solution **200** from overflowing, thereby preventing the solution from overflowing into another adjacent detection unit **100** and interfering with the detection result of another biological material. Therefore, compared with a general biochip, the biochip **10** of this embodiment may increase the volume of the solution **200** that the biochip **10** may hold through the arrangement of the fourth opening **O4**, so as to accommodate a larger volume of the solution **200** and improve the operational flexibility and convenience of the biochip **10**. In this way, multiple detection units **100** in the biochip **10** of this embodiment may be respectively used to detect different kinds of biological materials at the same time without the concern of overflow and subsequent cross-contamination between different detection units **100**. This in turn enables the biochip **10** to detect multiple biological materials at the same time.

[0047] Other embodiments are described below for illustrative purposes. It is to be noted that the following embodiments use the reference numerals and a part of the contents of the above embodiments, and the same reference numerals are used to denote the same or similar elements, and the description of the same technical contents is omitted. For the description of the omitted part, reference may be made to the above embodiments, and details are not described in the following embodiments.

[0048] FIG. **6** is a top schematic diagram of a detection unit in a biochip of another embodiment of the disclosure. Referring to FIG. **6** and FIG. **2C** at the same time, the biochip **10a** of this embodiment is similar to the biochip **10** in FIG. **2C**. The main difference between the two is that in the detection unit **100a** of the biochip **10a** of this embodiment, the semiconductor layer **120a** includes more reaction regions **121a**, source regions **122a**, and drain regions **123a**.

[0049] Specifically, referring to FIG. **6**, the semiconductor layer **120a** may include twelve reaction regions **121a**, six source regions **122a**, six drain regions **123a**, and one central opening **124**. Each pair of adjacent source regions **122a** and drain regions **123a** may occupy approximately 60 degrees of the entire 360-degree circular structure, and each source region **122a** (or drain region **123a**) may occupy approximately 30 degrees of the entire 360-degree circular structure. Thereby, the detection unit **100a** may further improve the detection sensitivity by accumulating signals detected by more reaction regions **121a** (or reaction units **1211**).

[0050] In some embodiments, more reaction units may also be added by adjusting the length of the reaction region (the length of the detection region) in the detection unit, thereby the detection unit may further improve detection sensitivity by accumulating signals detected by more reaction units.

[0051] To sum up, in the biochip and the manufacturing method thereof according to an embodiment of the disclosure, by making the diameter (or width) of the adding region larger than the diameter of the liquid bead, the solution may be prevented from overflowing due to the offset (or misalignment) of the adding position during addition. By designing the outline shape of the adding region to be circular to match the shape of the liquid beads, the surface tension of the liquid may be used to prevent the solution from overflowing when the amount of liquid in the solution is too much and is close to overflowing. In other words, this embodiment utilizes the “self-confined structure” formed by the wiring of the biochip itself to increase the alignment error tolerance space and accommodate more solution. By disposing multiple reaction regions in one detection unit to detect the same kind of biological material, the detection sensitivity may be improved by accumulating signals detected by the multiple reaction regions. Compared with general biochips, the biochip of this embodiment may increase the volume of solution that the biochip may hold through the arrangement of the fourth opening, so as to accommodate a larger volume of solution without the concern of overflow and subsequent cross-contamination. This in turn enhances the operational flexibility and convenience of the biochip. In addition, since the multiple detection units in the biochip of this embodiment may be respectively used to detect different kinds of

biological materials without the concern of overflow and subsequent cross-contamination between different detection units. This in turn enables the biochip to detect multiple biological materials at the same time without interfering with each other. In addition, in some embodiments, by increasing the number of reaction regions in the detection unit or adjusting the length of the reaction region in the detection unit (the length of the detection region) to increase the number of reaction units, the detection unit may further improve detection sensitivity by accumulating signals detected by more reaction units.

[0052] Although the disclosure has been described in detail with reference to the above embodiments, they are not intended to limit the disclosure. Those skilled in the art should understand that it is possible to make changes and modifications without departing from the spirit and scope of the disclosure. Therefore, the protection scope of the disclosure shall be defined by the following claims.

Claims

1. A biochip, configured for detecting biological material in a solution, comprising at least one detection unit, wherein the detection unit comprises: a substrate; a first insulation layer, disposed on the substrate; a semiconductor layer, disposed on the first insulation layer and comprises a plurality of reaction regions; a second insulation layer, disposed on the semiconductor layer and comprises: a first part; a second part, surrounding the first part; and a first opening, separating the first part and the second part and exposing the reaction regions; a first metal layer, disposed on the second part and comprising a source electrode, a drain electrode, a gate electrode, and a first wall structure that are separated from each other; a second metal layer, comprising: a second wall structure, disposed on the first wall structure; and a biological electrode, disposed on the first part; and a protection layer, disposed on the first metal layer and the second metal layer, having a second opening, a third opening and a fourth opening, and comprising: a flat part, covering the second part exposed by the first metal layer, and surrounding and defining the second opening; a first protruding part, covering the source electrode, the drain electrode, and the gate electrode, and surrounding and defining the third opening; and a second protruding part, covering the second wall structure, and surrounding and defining the fourth opening; wherein in a normal direction of the substrate, the first opening, the second opening, the third opening, and the fourth opening overlap.
2. The biochip according to claim 1, wherein the semiconductor layer further comprises a source region and a drain region, the source electrode surrounds the first opening and is electrically connected to the source region, the drain electrode surrounds the source electrode and is electrically connected to the drain region, the gate electrode surrounds the drain electrode and is electrically connected to the substrate, and the first wall structure surrounds the gate electrode.
3. The biochip according to claim 2, wherein the source electrode, the drain electrode, the gate electrode and the first wall structure are on a same layer, and the first wall structure electrically insulates the source electrode, the drain electrode, and the gate electrode.
4. The biochip according to claim 1, wherein the solution is disposed in at least the first opening, the second opening and the third opening, and the solution contacts the biological electrode and the reaction regions of the semiconductor layer.
5. The biochip according to claim 4, wherein the biological electrode and the second wall structure are on a same layer, the biological electrode and the second wall structure are separated from each other, and the second wall structure electrically insulates the biological electrode.
6. The biochip according to claim 1, wherein the first protruding part completely surrounds the third opening, and the second protruding part completely surrounds the fourth opening.
7. The biochip according to claim 1, wherein in the normal direction, an upper surface of the second protruding part is higher than an upper surface of the first protruding part, and the upper surface of the first protruding part is higher than an upper surface of the flat part.

8. The biochip according to claim 1, wherein the third opening is larger than the second opening, and the fourth opening is larger than the third opening.

9. The biochip according to claim 1, wherein the third opening comprises: an adding region, disposed corresponding to the first part; and a plurality of detection regions, connected to the adding region and disposed corresponding to the reaction regions.

10. A manufacturing method of a biochip, comprising: providing a substrate; forming a first insulation layer on the substrate; forming a semiconductor layer on the first insulation layer, wherein the semiconductor layer comprises a plurality of reaction regions; forming a second insulation layer on the semiconductor layer, wherein the second insulation layer comprises: a first part; a second part, surrounding the first part; and a first opening, separating the first part and the second part and exposing the reaction regions; forming a first metal layer on the second part, wherein the first metal layer comprises a source electrode, a drain electrode, a gate electrode, and a first wall structure that are separated from each other; forming a second metal layer, wherein the second metal layer comprises: a second wall structure, disposed on the first wall structure; and a biological electrode, disposed on the first part; and forming a protection layer on the first metal layer and the second metal layer, wherein the protection layer has a second opening, a third opening and a fourth opening, and comprises: a flat part, covering the second part exposed by the first metal layer, and surrounding and defining the second opening; a first protruding part, covering the source electrode, the drain electrode, and the gate electrode, and surrounding and defining the third opening; and a second protruding part, covering the second wall structure, and surrounding and defining the fourth opening; wherein in a normal direction of the substrate, the first opening, the second opening, the third opening, and the fourth opening overlap.
